Python Functions

BASIC FUNCTION EXAMPLES (1–10)

1. Simple function

```
def greet():
    print("Hello, Python!")
greet()
```

2. Function with parameters

```
def greet_user(name):
    print("Hello,", name)
greet_user("Raj")
```

3. Function with return value

```
def add(a, b):
    return a + b
print(add(10, 20))
```

4. Even or odd check

```
def is_even(num):
    return num % 2 == 0
print(is_even(6))
```

5. Square of a number

```
def square(n):
    return n * n
print(square(5))
```

6. Find maximum of two numbers

```
def maximum(a, b):
    return a if a > b else b
print(maximum(10, 25))
```

7. Default argument

```
def country(name="India"):
    print("Country:", name)
country()
country("USA")
```

8. Area of a circle

```
def circle_area(radius):
    return 3.14 * radius * radius
print(circle_area(7))
```

9. String length

```
def string_length(text):
    return len(text)
print(string_length("Python"))
```

10. Sum of list elements

```
def list_sum(lst):
    return sum(lst)
print(list_sum([1, 2, 3, 4]))
```

◈ INTERMEDIATE FUNCTION EXAMPLES (11–20)

11. Factorial

```
def factorial(n):
    result = 1
    for i in range(1, n+1):
        result *= i
    return result
print(factorial(5))
```

12. Prime number check

```
def is_prime(n):
    if n <= 1:
```

```
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True
print(is_prime(7))
```

13. Fibonacci series

```
def fibonacci(n):
    a, b = 0, 1
    for _ in range(n):
        print(a, end=" ")
        a, b = b, a + b
fibonacci(5)
```

14. Count vowels

```
def count_vowels(text):
    vowels = "aeiouAEIOU"
    return sum(1 for ch in text if ch in vowels)
print(count_vowels("Python"))
```

15. Reverse string

```
def reverse_string(s):
    return s[::-1]
print(reverse_string("Hello"))
```

16. Palindrome check

```
def is_palindrome(word):
return word == word[::-1]
print(is_palindrome("madam"))
```

17. List square using function

```
def square_list(lst):
    return [x*x for x in lst]
print(square_list([1, 2, 3]))
```

18. Keyword arguments

```
def student(name, age):
    print(name, age)
student(age=20, name="Amit")
```

19. Function returning multiple values

```
def math_ops(a, b):
    return a+b, a-b, a*b
print(math_ops(10, 5))
```

20. Nested function

```
def outer():
    def inner():
        return "Inside Inner Function"
    return inner()
print(outer())
```

◈ ADVANCED FUNCTION EXAMPLES (21–30)

21. *args example

```
def add_all(*args):
    return sum(args)
print(add_all(1, 2, 3, 4))
```

22. **kwargs example

```
def employee(**details):
    return details
print(employee(name="Raj", role="Trainer"))
```

23. Lambda function

```
square = lambda x: x * x
print(square(6))
```

24. Map with function

```
def double(x):
    return x * 2
print(list(map(double, [1, 2, 3])))
```

25. Filter with function

```
def is_even(n):
    return n % 2 == 0
print(list(filter(is_even, [1, 2, 3, 4])))
```

26. Reduce function

```
from functools import reduce

def add(a, b):
    return a + b
print(reduce(add, [1, 2, 3, 4]))
```

27. Recursive function

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n-1)

print(factorial(5))
```

28. Generator function

```
def count_up(n):
    for i in range(1, n+1):
```

```
        yield i
for num in count_up(5):
    print(num)
```

29. Function as argument

```
def shout(text):
    return text.upper()
def speak(func):
    return func("hello")
print(speak(shout))
```

30. Decorator example

```
def my_decorator(func):
    def wrapper():
        print("Before function")
        func()
        print("After function")
    return wrapper
@my_decorator
def say_hi():
    print("Hi!")
say_hi()
```